

# The Integrated Toroidal–Syntropic Model: Core Architecture, Conditional Weak-Field Limit, and Open Research Gates

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**Research-stage status.** This document reconstructs a coherent ITSM field hierarchy and records both successful and failed research gates. It is not a completed alternative to  $\Lambda$ CDM and does not claim that the open matching, screening, lensing, disk, perturbation, or driven-anisotropy problems have been solved.

## Abstract

The Integrated Toroidal–Syntropic Model (ITSM) is reconstructed as a layered research programme rather than a single scalar interpolation formula. The minimal canonical condensate has a stable finite-density branch, but explicit amplitude integration proves that its leading static-gradient response is quadratic, so UVIR-001 rejects it as the origin of the required spatial  $Y^{3/2}$  operator. UVIR-002 provisionally separates the complex condensate’s density, circulation and topology roles from a preferred-frame force scalar. UVIR-003 Stage A supplies a complete two-derivative frame basis and a regulated force truncation; its flat decoupling checks pass, while a verified Einstein–aether substitution supplies exact frame-sector mode speeds. The zero-gradient force block also factorizes at quadratic order into one nonnegative  $z = 2$  scalar, but its normalization remains unmatched. The full gate remains open: the force dispersion has unresolved metric-light-cone crossings,  $K_Q$  is unmatched, and the available longitudinal IR NDA scale is not a physical EFT cutoff. The conditional cubic force action still yields a square-root acceleration in a quasi-static high-symmetry limit, and the validated CBR-001 calculations establish topology-dependent Casimir pressure while finding  $H_t/H_p = 13/12$  only as a tunable transient in the tested free-field model. Strong phenomenology therefore remains downstream of the explicit UVIR, matter, screening, lensing, disk and cosmological gates.

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# 1 Scope, claim status, and reconstruction rule

The preceding ITSM manuscript combined an active-superfluid ontology, a Born–Infeld interpolation, an algebraic galactic force, an open-system source and multiple cosmological claims. Subsequent equation audits and numerical gates showed that these ingredients do not yet follow from one action. The purpose of this alpha is therefore architectural: it states what the model is, which results are presently supportable, and what must still be calculated.

Four labels are used throughout:

**Derived.** A checked result that follows from declared assumptions within its stated domain.

**Conditional.** A result that also requires a named postulate, matching hypothesis, background choice or symmetry limit.

**Open.** A well-defined derivation or numerical test remains incomplete.

**Rejected.** The stated derivation or implementation failed and is not live support for the framework.

These labels apply to derivations rather than to broad themes. For example, topology-dependent Casimir stress is derived for the declared periodic scalar calculation, while a universal 13/12 expansion ratio from cycle counting is rejected. A rejected shortcut does not reject  $T^3$  topology; likewise, a successful local effective action does not establish a relativistic or cosmological completion.

## 1.1 Conventions

We use metric signature  $(-, +, +, +)$  and natural units  $c = \hbar = 1$  except where factors are restored explicitly, as in the Casimir lattice energy. The reduced Planck mass and gravitational coupling satisfy

$$M_{\text{P}}^{-2} = 8\pi G, \quad \kappa = 8\pi G. \quad (1)$$

These conventions make the covariant  $Y^{3/2}$  coefficient and the static cubic-gradient coefficient directly comparable.

Table 1: High-level status at the start of the core recovery.

| Component                                          | Status      | Boundary of the result                         |
|----------------------------------------------------|-------------|------------------------------------------------|
| Finite-density plenum                              | Conditional | Physical postulate; microscopic action open    |
| Compact $T^3$ topology                             | Conditional | Adopted boundary condition; scale open         |
| $Y^{3/2}$ square-root branch                       | Conditional | Local, quasi-static, high symmetry             |
| Present Born–Infeld origin                         | Rejected    | Does not give the required IR response         |
| $C_{\text{proj}} = 2/3$ local vertex               | Conditional | Geometric motivation; matching open            |
| Rectangular- $T^3$ Casimir tensor                  | Derived     | Periodic free scalar with stated moduli        |
| Persistent 13/12 from free-scalar Casimir stress   | Rejected    | CBR-001 finds only transient crossings         |
| Persistent 13/12 from a derived driven ITSM sector | Open        | Requires TOP/WAK/reservoir closure and CBR-002 |
| Screening, PPN and lensing                         | Open        | Relativistic completion absent                 |
| Realistic disk field solution                      | Open        | Periodic nonlinear PDE required                |
| Driven reservoir anisotropy                        | Open        | Constitutive conversion not derived            |

The reconstruction rule is that no observational claim may have a stronger status than its upstream field sector. Historical calculations remain useful as diagnostics and provenance, but they are not promoted into the core merely because their numerical values are suggestive.

## 2 Core fields and layered architecture

### 2.1 Metric, condensate, and finite-density frame

The gravitational variable is the spacetime metric  $g_{\mu\nu}$ . The candidate plenum order parameter is complex,

$$\Phi = \frac{\rho}{\sqrt{2}} e^{i\Theta}, \quad (2)$$

so that amplitude fluctuations, phase excitations, zeros of the order parameter and winding sectors can be represented within one microscopic language. UVIR-001 rejects the minimally kinetic analytic candidate as the direct origin of the spatial cubic force operator, but it does not remove these density, circulation, defect and topology roles.

A unit timelike vector  $U^\mu$  defines the preferred frame,

$$U^\mu U_\mu = -1, \quad h^{\mu\nu} = g^{\mu\nu} + U^\mu U^\nu. \quad (3)$$

UVIR-003 Stage A provisionally chooses  $U^\mu$  as an independent constrained dynamical field. The condensate current may be aligned with it through an interaction energy, but is not algebraically identified with it. This avoids double-counting a derived condensate velocity and an independent aether field. The choice remains provisional until the coupled constraint and stability gate closes.

## 2.2 Separate preferred-frame force scalar and matter

The low-acceleration force mediator is denoted  $\psi$ . Under the selected UVIR-002 architecture it is a separate preferred-frame scalar, not the phase of  $\Phi$  relabelled at low energy. Independent temporal and spatial invariants can therefore be formed from  $U^\mu \nabla_\mu \psi$  and  $h^{\mu\nu} \nabla_\mu \psi \nabla_\nu \psi$ . The cubic spatial operator is a conditional infrared EFT input; it has not been matched to a microscopic completion.

Matter fields are collected in  $\Psi_m$ . The matter–force interaction must be derived once, then used consistently in both the scalar source and the matter stress-energy exchange. No coefficient inferred from a convenient normalization is promoted to an observable coupling before MAT-001.

## 2.3 Topology, reservoir, and non-equilibrium sectors

Spatial slices are assigned compact  $T^3$  boundary conditions with lengths  $L_i$ . Their ratios are shape moduli. The global topology controls allowed modes and winding; it does not directly fix every local Wilson coefficient.

An explicit reservoir sector  $R$  permits the observable matter–plenum system to be open while the complete covariant system remains conserved. A wake or memory variable  $W$  and a shape sector are permitted research sectors, not assumed completed degrees of freedom.

The schematic inventory is

$$S_{\text{complete}} = S_{\text{EH}} + S_\Phi + S_U + S_{\text{align}} + S_\psi + S_m + S_R + S_{\text{int}} + S_W, \quad (4)$$

where  $S_W$  is included only if a causal and stable wake description is found. The displayed sectors define a research architecture, not a completed microscopic action. Effective operators must be matched and overlapping UV and IR descriptions must not be counted as independent copies of the same degree of freedom.

## 2.4 Architecture versus prediction

Equation (4) is a sector map. In particular,

$$\text{energy throughput} \not\Rightarrow \text{anisotropic pressure} \quad (5)$$

without a shape-dependent conversion law, and

$$\text{compact topology} \not\Rightarrow C_{\text{proj}} = \frac{2}{3} \quad \text{or} \quad \frac{H_t}{H_p} = \frac{13}{12} \quad (6)$$

without separate matching and backreaction calculations. Likewise, a positive frozen-background dispersion is not a proof of coupled stability, causality or weak coupling.

### 3 Ultraviolet–infrared hierarchy

#### 3.1 UVIR-001: the minimal condensate route closes negative

The tested microscopic candidate is

$$S_\Phi = \int d^4x \sqrt{-g} \left[ -\nabla_\mu \Phi^* \nabla^\mu \Phi - m^2 |\Phi|^2 - \frac{\lambda_4}{2} |\Phi|^4 - \frac{\lambda_6}{3\Lambda^2} |\Phi|^6 \right]. \quad (7)$$

Writing  $\Phi = \rho e^{i\Theta}/\sqrt{2}$  and  $Z = -\nabla_\mu \Theta \nabla^\mu \Theta$ , its stable finite-density branch can be reduced to a phase action  $P(Z)$ . In the pure-sextic limit the explicit amplitude integration gives

$$P(Z) = \frac{2\Lambda}{3\sqrt{\lambda_6}} (Z - m^2)^{3/2}. \quad (8)$$

This nonanalytic power does not supply the required static spatial operator. About a timelike background  $Z_0 = \mu^2$ , a static phase gradient  $q = |\nabla \vartheta|$  instead yields

$$P(\mu^2 - q^2) - P(\mu^2) = -P_Z(\mu^2) q^2 + \mathcal{O}(q^4), \quad P_Z = \frac{\rho_0^2}{2} > 0. \quad (9)$$

The leading response is therefore quadratic for every stable nonzero branch with the canonical phase kinetic term. UVIR-001 is closed negative for this minimal candidate. The result rejects neither the condensate’s density and topology roles nor a bottom-up spatial  $Y^{3/2}$  EFT.

#### 3.2 UVIR-002: provisional two-sector route

UVIR-002 also rejects a standalone nonrelativistic gradient-dominated branch: the branch producing the desired cubic spatial action has a negative quadratic time kinetic coefficient. The least-incomplete route tested separates the jobs of the condensate and force mediator. The independent invariants are

$$\mathcal{Q} = U^\mu \nabla_\mu \psi, \quad \mathcal{Y} = h^{\mu\nu} \nabla_\mu \psi \nabla_\nu \psi. \quad (10)$$

Here  $\Phi$  retains finite density, circulation, defects and topology, while  $(\psi, U^\mu)$  carries the conditional low-acceleration response. This is a provisional architecture selection, not microscopic closure.

#### 3.3 UVIR-003 Stage A action and limited pass

Stage A chooses an independent unit vector and retains the complete Einstein–aether two-derivative basis,

$$\begin{aligned} \mathcal{L}_U = & -\frac{M_U^2}{2} [c_1 (\nabla_\mu U_\nu) (\nabla^\mu U^\nu) + c_2 (\nabla_\mu U^\mu)^2 \\ & + c_3 (\nabla_\mu U_\nu) (\nabla^\nu U^\mu) - c_4 a_\mu a^\mu] + \frac{\lambda}{2} (U^\mu U_\mu + 1), \end{aligned} \quad (11)$$

where  $a_\mu = U^\nu \nabla_\nu U_\mu$ . A positive alignment energy couples the frame to the condensate current without imposing equality. The regulated force truncation is

$$\mathcal{L}_\psi = \frac{K_Q}{2} Q^2 - A \mathcal{Y}^{3/2} - \frac{\gamma}{2M_*^2} (\Delta_U \psi)^2. \quad (12)$$

The symbolic Stage-A calculation gives the necessary frozen-metric conditions

$$c_{14} > 0, \quad c_1 > 0, \quad c_{123} > 0, \quad K_Q > 0, \quad A > 0, \quad \gamma > 0, \quad (13)$$

and at zero background gradient the regulator produces

$$\omega^2 = \frac{\gamma}{K_Q M_*^2} k^4. \quad (14)$$

This is a pass only in the declared flat, constant-frame decoupling limit.

### 3.4 Stage B frame-sector diagnostic

A verified convention map to the Einstein–aether review [3] maps all four  $c_i$  identically. Substitution of that source’s coupled metric–aether result gives

$$s_2^2 = \frac{1}{1 - c_{13}}, \quad (15)$$

$$s_1^2 = \frac{c_1 - \frac{1}{2}c_1^2 + \frac{1}{2}c_3^2}{c_{14}(1 - c_{13})}, \quad (16)$$

$$s_0^2 = \frac{c_{123}(2 - c_{14})}{c_{14}(1 - c_{13})(2 + c_{13} + 3c_2)}. \quad (17)$$

When all  $c_i$  are scaled uniformly to zero, the spin-1 and spin-0 expressions reduce to the Stage-A ratios  $c_1/c_{14}$  and  $c_{123}/c_{14}$ . This confirms that those ratios are weak-metric-coupling limits, not the full mode speeds. It is a checked literature substitution, not an independent ITSM-side linearized Einstein-equation derivation, and it excludes the coupled  $\Phi$ – $U$ – $\psi$  system.

### 3.5 Zero-gradient force-block factorization

On the declared background  $\bar{\psi} = \text{constant}$ , every derivative of the force field begins at first perturbative order. Consequently,

$$\mathcal{Q}^2 = \dot{\pi}^2 + \mathcal{O}(3), \quad \mathcal{Y}^{3/2} = \mathcal{O}(3), \quad (\Delta_U \psi)^2 = (\nabla^2 \pi)^2 + \mathcal{O}(3), \quad (18)$$

where metric and frame corrections enter only in the indicated higher-order terms. The force scalar therefore factorizes from the lapse, shift, frame and condensate blocks at quadratic order for the declared truncation:

$$S_\pi^{(2)} = \frac{1}{2} \int d^4x \left[ K_Q \dot{\pi}^2 - \frac{\gamma}{M_*^2} (\nabla^2 \pi)^2 \right]. \quad (19)$$

It carries one nonnegative  $z = 2$  scalar mode for  $K_Q, \gamma > 0$ . This bounded result does not reduce the remaining metric–frame–condensate block.

A constant field redefinition  $\psi_c = s\psi$  sends

$$K_Q \rightarrow \frac{K_Q}{s^2}, \quad A \rightarrow \frac{A}{s^3}, \quad \gamma \rightarrow \frac{\gamma}{s^2}, \quad C_m \rightarrow \frac{C_m}{s}, \quad q \rightarrow sq. \quad (20)$$

Thus  $\gamma/K_Q$ ,  $A/K_Q^{3/2}$ ,  $C_m/\sqrt{K_Q}$  and  $Aq/K_Q$  are invariant, whereas  $K_Q$  alone is not. UVIR-003 can determine a structural stability domain in invariant ratios, but a parent microscopic calculation or MAT-001 must fix the physical normalization and select a point in that domain. The earlier  $K_Q \sim M_P^2$  estimate remains a speculative dimensional analogy.

### 3.6 Causality and cutoff status

Around a nonzero spatial background gradient of magnitude  $q$ , the force fluctuation has

$$\omega^2 = \frac{3Aq(1 + \cos^2 \theta)}{K_Q} k^2 + \frac{\gamma}{K_Q M_*^2} k^4. \quad (21)$$

Thus the long-wavelength metric-light-cone threshold depends on the unmatched ratio  $Aq/K_Q$ , while phase and group velocities grow without bound at large  $k$  inside the truncated dispersion. Preferred-frame metric superluminality is not by itself a proof of acausality, but the foliation-causality test and the physical EFT domain have not been established.

Expansion of the cubic vertex supplies only the restricted, time-normalized longitudinal IR NDA scale

$$\Lambda_{\text{NDA, long}}^{(\text{IR})} \sim \frac{K_Q^{3/4}}{\sqrt{A}}. \quad (22)$$

It is not the physical cutoff  $\Lambda_{\text{EFT}}$ : transverse normalization,  $k^4$ -dominated power counting and a loop or unitarity analysis remain open. The comparison of the light-cone crossing scale with  $\Lambda_{\text{EFT}}$  therefore cannot yet be made.

A separate dimensional analogy,  $K_Q = k_Q M_P^2$ , would place the long-wavelength threshold at order  $a_0$  for order-unity coefficients. That exercise is explicitly speculative: it is introduced only as a priority signal for deriving  $K_Q$ , not as a property of ITSM.

### 3.7 Conditional infrared operator

Define the normalized spatial invariant

$$Y = \frac{h^{\mu\nu} \nabla_\mu \psi \nabla_\nu \psi}{a_0^2}. \quad (23)$$

The target low-acceleration operator remains

$$\mathcal{L}_{\psi, \text{IR}} = -\frac{2C_{\text{IR}}}{3} M_P^2 a_0^2 Y^{3/2} + \mathcal{O}(Y^2, \nabla^2 / \Lambda_{\text{EFT}}^2). \quad (24)$$

Neither  $C_{\text{IR}}$ ,  $K_Q$  nor the normalized matter vertex has been derived. The preceding analytic Born–Infeld expression also remains rejected as the source of this branch because its small-invariant equation is linear at leading order. UVIR-003 is therefore in progress and MAT-001 remains blocked.

## 4 Conservation and syntropic throughput

The observable matter–plenum subsystem may exchange energy and momentum with a reservoir sector external to that observable subsystem but contained within the complete conserved theory. The complete system must satisfy the contracted Bianchi identity. We therefore distinguish two currents:

$$\nabla_\mu T_m^{\mu\nu} = Q_{\text{mp}}^\nu, \quad (25)$$

$$\nabla_\mu T_P^{\mu\nu} = -Q_{\text{mp}}^\nu + Q_{\text{syn}}^\nu, \quad (26)$$

$$\nabla_\mu T_R^{\mu\nu} = -Q_{\text{syn}}^\nu. \quad (27)$$

Adding Eqs. (25)–(27) gives

$$\nabla_\mu (T_m^{\mu\nu} + T_P^{\mu\nu} + T_R^{\mu\nu}) = 0. \quad (28)$$

$Q_{\text{mp}}^\nu$  is the local matter–plenum exchange generated by the same interaction that sources the force scalar.  $Q_{\text{syn}}^\nu$  is reservoir throughput. Treating them as one undifferentiated  $Q^\nu$  hides two different physical jobs and prevents a clean local/cosmological split.

#### 4.1 Candidate local interaction

A universal conformal matter metric illustrates how a force and an exchange current can share an origin,

$$\tilde{g}_{\mu\nu} = e^{2C_m\psi/c^2} g_{\mu\nu}, \quad S_m = S_m[\tilde{g}_{\mu\nu}, \Psi_m]. \quad (29)$$

For pressureless matter its nonrelativistic limit contains

$$\mathcal{L}_{\text{int}}^{\text{NR}} = -C_m \rho_b \psi, \quad (30)$$

and its exchange current is proportional to the matter trace times  $\nabla^\nu \psi$ . This is a candidate matter coupling, not a complete relativistic phenomenology: a purely conformal rescaling does not by itself provide the required lensing.

#### 4.2 Reservoir throughput and anisotropic stress

In the plenum rest frame a general stress tensor includes

$$T_P^{\mu\nu} = \rho_P U^\mu U^\nu + p_P h^{\mu\nu} + \pi^{\mu\nu}. \quad (31)$$

The traceless spatial tensor  $\pi^{\mu\nu}$ , not the energy-transfer vector alone, carries directional stress. In a biaxial state one may define

$$\Pi = p_t - p_p, \quad \pi^i_j = \text{diag} \left( -\frac{2\Pi}{3}, \frac{\Pi}{3}, \frac{\Pi}{3} \right). \quad (32)$$

For the biaxial expansion used below, the shear tensor is

$$\sigma^i_j = \text{diag} \left( -\frac{2\delta}{3}, \frac{\delta}{3}, \frac{\delta}{3} \right), \quad \pi^{\mu\nu} \sigma_{\mu\nu} = \frac{2}{3} \Pi \delta. \quad (33)$$

The last equality is the exact anisotropic work term for these conventions. A relation such as  $Q_{\text{syn}}^\nu \propto \Theta U^\nu$  injects energy along the background flow but does not determine  $\Pi$ . A geometry-dependent action or constitutive law is required to convert throughput into shear.

No fixed redshift law for  $Q_{\text{syn}}^\nu$  is adopted in this alpha. Its background and perturbation dependence must follow from the reservoir model, and local decoupling must be tested rather than asserted from  $\Theta \rightarrow 0$ .

### 5 Conditional low-acceleration matter coupling

#### 5.1 Inverse construction of the cubic-gradient operator

Nonlinear Poisson equations derived from nonquadratic potential actions have a standard precedent in AQUAL-type theories [4]. Here we isolate the coefficient normalization required by the ITSM infrared branch. Let  $q = |\nabla \psi|$  and consider the static force-scalar action

$$S_\psi = \int dt d^3x [-K(q) - C_m \rho_b \psi]. \quad (34)$$



Variation yields

$$\nabla \cdot \left[ \frac{K'(q)}{q} \nabla \psi \right] = C_m \rho_b. \quad (35)$$

The required cubic-gradient operator is parameterized as

$$K(q) = \frac{C_{\text{IR}}}{12\pi G a_0} q^3, \quad C_{\text{IR}} > 0. \quad (36)$$

The complete static weak-field action in this normalization is

$$S_{\text{WF}} = \int dt d^3x \left[ -\frac{|\nabla \Phi_N|^2}{8\pi G} - \rho_b \Phi_N - \frac{C_{\text{IR}}}{12\pi G a_0} |\nabla \psi|^3 - C_m \rho_b \psi \right]. \quad (37)$$

For independent positive coefficients  $C_m$  and  $C_{\text{IR}}$ , the sourced equation is

$$\nabla \cdot \left[ \frac{|\nabla \psi|}{a_0} \nabla \psi \right] = 4\pi G \frac{C_m}{C_{\text{IR}}} \rho_b. \quad (38)$$

For spherical symmetry this gives

$$|\nabla \psi|^2 = \frac{C_m}{C_{\text{IR}}} a_0 g_N. \quad (39)$$

Since the physical acceleration is  $g_P = C_m |\nabla \psi|$ , the invariant observable square-root coefficient is

$$g_P = C_{\text{obs}} \sqrt{a_0 g_N}, \quad \boxed{C_{\text{obs}} = \frac{C_m^{3/2}}{\sqrt{C_{\text{IR}}}}}. \quad (40)$$

Only under the normalization choice  $C_m = C_{\text{IR}} = C$  does  $C_{\text{obs}} = C$ . That convenient choice is not a physical derivation of  $C = 2/3$ ; MAT-001 must calculate the invariant combination in Eq. (40).

## 5.2 Compact toroidal source

On compact  $T^3$  there is no boundary at spatial infinity, and a periodic divergence integrates to zero. Write

$$\rho_b = \bar{\rho}_b + \delta \rho_b, \quad \psi = \bar{\psi}(t) + \varphi(t, \mathbf{x}), \quad (41)$$

with zero spatial averages for  $\delta \rho_b$  and  $\varphi$ . The local periodic equation has the form

$$D_i \left[ \frac{|D\varphi|}{a_0} D^i \varphi \right] = 4\pi G \frac{C_m}{C_{\text{IR}}} \delta \rho_b + \mathcal{S}_Q, \quad \int_{T^3} \left( 4\pi G \frac{C_m}{C_{\text{IR}}} \delta \rho_b + \mathcal{S}_Q \right) dV = 0. \quad (42)$$

Here  $\mathcal{S}_Q \equiv \mathcal{S}_Q[\delta Q_{\text{syn}}]$  is a conditional zero-mean scalar source representing the inhomogeneous projection of reservoir–plenum exchange into the force-scalar equation. Its functional map from the vector exchange current has not been derived; setting  $\mathcal{S}_Q = 0$  is the local-static baseline rather than a claim that reservoir perturbations vanish identically. The homogeneous mode belongs to the cosmological/open-system problem. A localized spherical solution is therefore the small-region, compensated-overdensity limit of Eq. (42), not an isolated global cosmology.

### 5.3 Disk geometry and coefficient status

For a general density distribution, equality of divergences permits

$$\frac{|\nabla\varphi|}{a_0}\nabla\varphi = \frac{C_m}{C_{\text{IR}}}\nabla\Phi_N + \nabla \times \mathbf{H}. \quad (43)$$

The solenoidal field vanishes in sufficient symmetry but is generally nonzero for a finite disk. Consequently a pointwise square-root prescription is a historical algebraic approximation. DISK-001 must solve Eq. (42) for realistic baryonic sources before  $C_{\text{obs}} = 2/3$  and  $C_{\text{obs}} \simeq 1$  can be fairly compared.

The separate coefficients change under a rescaling of  $\psi$ , while the physical response is carried by the invariant combination  $C_{\text{obs}} = C_m^{3/2}/\sqrt{C_{\text{IR}}}$ . The trace ratio  $2/3$  remains a geometric motivation and a matching hypothesis, not a completed local Wilson coefficient.

## 6 Toroidal topology and validated Casimir backreaction

### 6.1 Periodic rectangular lattice

For a massless scalar with periodic lengths  $(L_1, L_2, L_3)$ , the renormalized energy density used in CBR-001 is the rectangular-periodic Epstein-lattice result developed in Ref. [5]:

$$\rho_{\text{Cas}} = -\frac{\hbar c}{2\pi^2} \sum_{\mathbf{n} \in \mathbb{Z}^3 \setminus \{0\}} \frac{1}{[(n_1 L_1)^2 + (n_2 L_2)^2 + (n_3 L_3)^2]^2}. \quad (44)$$

The directional pressures follow thermodynamically,

$$p_i = -\frac{1}{L_j L_k} \frac{\partial E_{\text{Cas}}}{\partial L_i}, \quad E_{\text{Cas}} = L_1 L_2 L_3 \rho_{\text{Cas}}. \quad (45)$$

The Stage-1 and Stage-2 calculations validate cutoff convergence, cubic isotropy, pressure derivatives, the massless trace relation, dimensional scaling and the sign change of  $p_t - p_p$  across the cubic shape. Thus compact shape-dependent anisotropic stress is a concrete mechanism. It is not a universal cycle-counting coefficient. A recent direct semiclassical calculation likewise derives a finite topology-dependent stress tensor on  $T^3$  and its anisotropic de Sitter backreaction [8].

### 6.2 Biaxial backreaction

For

$$ds^2 = -dt^2 + a_p^2 dx^2 + a_t^2 (dy^2 + dz^2), \quad (46)$$

define

$$H = \frac{H_p + 2H_t}{3}, \quad \delta = H_t - H_p, \quad r = \frac{a_t}{a_p}. \quad (47)$$

Then

$$\dot{\delta} + 3H\delta = \kappa(p_t - p_p), \quad \dot{r} = r\delta, \quad (48)$$

with the Hamiltonian constraint

$$3H^2 - \frac{\delta^2}{3} = \kappa\rho_{\text{total}}. \quad (49)$$

These are the biaxial specialization of the standard Bianchi-I system; see, for example, Ref. [6]. The Stage-3A solver reproduces the perturbative solution and preserves the constraint and Casimir continuity equation to substantially better than the specified tolerances.

### 6.3 The Stage-3B ratio test

Let

$$x = \frac{\delta}{H}, \quad q = \frac{H_t}{H_p} = \frac{1 + x/3}{1 - 2x/3}. \quad (50)$$

The historical target  $q = 13/12$  corresponds to

$$x_\star = \frac{3}{38}. \quad (51)$$

The small-source fixed-background solution is

$$x(N) = S (e^{-3N} - e^{-4N}), \quad (52)$$

with maximum  $(27/256)S$  at  $N = \ln(4/3)$ . Merely reaching  $x_\star$  therefore requires

$$S \geq \frac{128}{171} \simeq 0.748538, \quad (53)$$

already outside the strict small-perturbation regime.

The nonlinear root search finds target-reaching amplitudes for five of the tested initial shapes. Every case is classified as a transient crossing, with only approximately 0.27–0.30 e-folds within one percent of the target. Four thresholds are nonperturbative and one is marginal by the stated Casimir-fraction diagnostic. Every valid candidate returns to  $q = 1$  by  $N = 20$ ; no quasi-plateau or attractor is found.

There is also a kinematic warning. Since

$$\frac{d \ln r}{dN} = x, \quad (54)$$

a constant  $q \neq 1$  implies continuous evolution of the shape ratio. It cannot be a fixed torus shape. A persistent anisotropic state would have to be a driven solution, not a static geometric ratio.

For a desired trajectory  $x(N)$ , the required anisotropic stress is

$$\Pi_{\text{req}} = \frac{H^2}{\kappa} \left[ \frac{dx}{dN} + x \left( 3 + \frac{d \ln H}{dN} \right) \right]. \quad (55)$$

In de Sitter with constant  $x = 3/38$  this becomes

$$\Pi_{\text{req}} = \frac{9H^2}{38\kappa}. \quad (56)$$

For the CBR-001 physical lengths  $L_p = L_\star a r^{-2/3}$  and  $L_t = L_\star a r^{1/3}$ , the anisotropic Casimir source is more precisely

$$\Pi_{\text{Cas}}(a, r) = \frac{\hbar c}{L_\star^4} a^{-4} r^{8/3} [\hat{p}_t(r) - \hat{p}_p(r)] \equiv \frac{\hbar c}{L_\star^4} a^{-4} F_\Pi(r). \quad (57)$$

Thus the explicit scale dependence is radiation-like, while the shape function evolves whenever  $r$  evolves. In every validated free-field trajectory the source nevertheless becomes negligible and cannot maintain Eq. (55). A future CBR-002 test must derive any additional shape, driven or memory stress without inserting the target ratio into a coefficient.

## 7 Wake, circulation, and non-equilibrium hypotheses

The original ITSM described baryonic matter as exciting a torsional or acoustic wake in a rotational medium. The static equation Eq. (42) has no independent memory: for specified boundary data it tracks the instantaneous source. A persistent, advected or detached wake therefore requires additional dynamics.

A schematic relaxation equation is

$$\tau_W U^\mu \nabla_\mu W + W = \mathcal{F}[\psi, \rho_b, r, Q_{\text{syn}}], \quad (58)$$

or, if wave propagation is essential, a hyperbolic field equation with a second time derivative. Equation (58) is not adopted as a law. It identifies the quantities that WAK-001 must derive or bound. Its first-order form is motivated only as a causal-relaxation template of the Israel–Stewart type [7]; it is not a derivation of an ITSM constitutive law.

A viable wake sector must satisfy all of the following:

1. a well-posed initial-value problem;
2. causal characteristic speeds in the physical frame;
3. a positive or bounded energy accounting;
4. a controlled static limit reproducing the IR field equation;
5. explicit momentum exchange with matter and the reservoir;
6. decay or transport laws rather than indefinite unsupported memory.

Similarly, circulation is represented naturally by the phase in Eq. (2),

$$\oint \nabla_i \Theta dx^i = 2\pi n, \quad n \in \mathbb{Z}, \quad (59)$$

where zeros of  $\rho$  permit defect cores. The relationship between these microscopic winding sectors and a global toroidal current is the VOR-001 gate. No black-hole, lensing or galactic coefficient follows from the winding number without further dynamics.

For driven anisotropy one may organize the bookkeeping as

$$\Pi_{\text{total}} = \Pi_{\text{Cas}} + \Pi_{\text{shape}} + \Pi_{\text{driven}} + \Pi_{\text{memory}}. \quad (60)$$

Only  $\Pi_{\text{Cas}}$  has been calculated in the present programme. The other terms are labels for research sectors. Writing Eq. (60) does not establish that they exist or have the required sign and magnitude.

## 8 Cosmological architecture and current limits

The homogeneous condensate, reservoir and topology moduli must define one fiducial cosmological background before CMB, growth, age or distance results are calculated. Earlier analyses used several dataset-conditioned values of the syntropic exponent or effective equation of state and sometimes mixed them across diagnostics. Those outputs remain provenance and cannot be treated as one canonical cosmology.

At background level, a conservative sector decomposition has the form

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \pi^{\mu\nu}\sigma_{\mu\nu} = Q, \quad (61)$$

$$\dot{\rho}_R + 3H(\rho_R + p_R) = -Q, \quad (62)$$

For the biaxial conventions of Sec. 6, Eq. (33) gives the exact specialization

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \frac{2}{3}\Pi\delta = Q. \quad (63)$$

The total continuity equation follows by summing the sectors.

No predetermined law such as  $Q \propto (1+z)^{-3}$  is adopted here. In particular,  $(1+z)^{-3} = a^3$  grows toward late time and cannot simultaneously be described as an early-universe enhancement. COS-001 must derive the source and fit one background with an explicit parameter inventory.

## 8.1 What CBR-001 changes

CBR-001 does not remove topology-dependent cosmological stress. It shows that the tested passive free-field source has the evolving form  $\Pi_{\text{Cas}} = a^{-4}F_{\Pi}(r)$  and becomes unable to support a persistent 13/12 directional expansion ratio. Therefore

- $H_0 = 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$  is not a parameter-free ITSM prediction;
- assigning Planck and SH0ES to opposite directional extrema is not a derived observation model;
- any future directional expansion must be passed through actual survey window functions;
- a driven anisotropic state requires a derived source beyond free Casimir stress.

## 8.2 Perturbations

PERT-001 must vary the same total action and exchange currents used for the background. Effective CAMB or CLASS mappings are useful diagnostics, but they do not replace the derivation of scalar, vector and tensor perturbations, initial conditions, stability and observable transfer functions. Until that gate closes, CMB-peak and  $S_8$  values are diagnostic calculations rather than canonical predictions.

## 9 Dependency-ordered closure gates

**Table 2: Research gates required before restoration of strong predictions.**

| Gate and status                       | Question                                                  | Pass condition                                                          |
|---------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------|
| UVIR-001<br><i>Closed negative</i>    | Does the minimal condensate yield spatial $Y^{3/2}$ ?     | Candidate rejected; replacement route required                          |
| UVIR-002<br><i>Closed provisional</i> | Which replacement architecture survives initial checks?   | Two-sector preferred-frame route selected for testing                   |
| UVIR-003<br><i>In progress</i>        | Is the selected action stable, causal and weakly coupled? | Full coupled constraint reduction, physical cutoff and background tests |
| MAT-001<br><i>Blocked</i>             | What is the normalized matter vertex?                     | Force and exchange from one action; $C_m$ , $C_{IR}$ and $K_Q$ matched  |
| SCR-001<br><i>Open</i>                | Is the force screened at high gradient?                   | Ephemeris and binary constraints satisfied                              |
| LEN-001<br><i>Open</i>                | What metric do matter and light see?                      | PPN, lensing and wave propagation derived                               |
| DISK-001<br><i>Open</i>               | What is the exact disk response?                          | Validated periodic nonlinear solver and curl field                      |
| TOP-001<br><i>Open</i>                | How do shape moduli evolve?                               | Covariant moduli action and stable dynamics                             |
| VOR-001<br><i>Open</i>                | How is global circulation carried?                        | Winding and defect sector with finite energy                            |
| WAK-001<br><i>Open</i>                | Can wakes retain causal memory?                           | Hyperbolic or relaxation model with energy accounting                   |
| CBR-002<br><i>Open</i>                | Can derived sectors sustain anisotropy?                   | Untuned non-equilibrium solution and stability test                     |
| COS-001<br><i>Open</i>                | What is the fiducial background?                          | One coherent fit from declared equations                                |
| PERT-001<br><i>Open</i>               | What are the perturbation equations?                      | Action-derived stable transfer system                                   |

The dependency order is

$$\begin{aligned}
& \text{UVIR-001} \rightarrow \text{UVIR-002} \rightarrow \text{UVIR-003} \rightarrow \text{MAT-001} \\
& \rightarrow (\text{SCR}, \text{LEN}) \rightarrow \text{DISK} \\
& \rightarrow (\text{TOP}, \text{VOR}, \text{WAK}, R) \rightarrow \text{CBR-002} \\
& \rightarrow (\text{COS}, \text{PERT}).
\end{aligned} \tag{64}$$

The frame-sector literature substitution, zero-gradient force-block factorization and longitudinal IR NDA estimate are progress within UVIR-003, not new closed gates. The force block contains one

nonnegative  $z = 2$  scalar in the declared zero-gradient truncation, while a rescaling audit shows that a standalone  $K_Q$  is not identifiable before physical field normalization. The physical EFT cutoff, nonzero-gradient mixing and the full metric–frame–condensate constraint system remain unresolved. Likewise, running CBR-002 with a source adjusted to hold 13/12 would test a tuning procedure, not ITSM, and fitting the algebraic disk law does not answer the periodic nonlinear field problem.

## 10 Falsifiability without overclaiming

The recovery replaces spectacular but weakly derived forecasts with a sequence of gate-level falsifiers.

### 10.1 Near-term theoretical falsifiers

1. UVIR-001 has already rejected the minimally kinetic condensate as the direct origin of the spatial cubic law. If UVIR-003 cannot produce a stable, causal and weakly coupled two-sector completion with a controlled physical cutoff, the selected replacement architecture is rejected as well.
2. If no matching calculation fixes  $K_Q$ ,  $C_{\text{IR}}$  and the normalized matter vertex in a domain compatible with the mode constraints, MAT-001 cannot open and the local force law remains an uncompleted EFT ansatz.
3. If a normalized matter vertex cannot satisfy galactic and local bounds within one action, MAT-001/SCR-001 reject the universal coupling architecture.
4. If no relativistic completion produces acceptable PPN and lensing potentials, the local acceleration law cannot serve as a gravitational model.
5. If the periodic disk solution preserves the empirical tension of the algebraic  $C_{\text{obs}} = 2/3$  implementation, disk geometry cannot rescue that matching hypothesis.
6. If derived reservoir, shape and wake sectors have no stable driven anisotropic solution, CBR-002 rejects persistent topological Hubble anisotropy within that architecture.

### 10.2 Observational claims held downstream

NANOGrav spectral features require a derived eigenmode spectrum and detector response. Cluster offsets require causal wake dynamics, a collision calculation and a lensing metric. Stellar-population claims require an independent star-formation model. CMB and growth require a coherent background and perturbation theory. None is presented here as a clean framework-level falsifier.

### 10.3 Reproducibility rule

Every future numerical claim must identify its source equations, code entry point, data version, parameter inventory, validation controls and a machine-readable summary. A reported statistic must be explicitly computed by the cited pipeline. Negative and partial gate results remain part of the model record rather than being hidden by a new ansatz.

## Conclusion

The recovery preserves the defining ITSM idea—a driven, finite-density, rotational medium with compact topology and sector exchange—while recording where candidate mechanisms fail. The minimal condensate route to the spatial cubic law is closed negative; the replacement preferred-frame force sector has passed only bounded Stage-A and frame-subsystem checks. Its coupled stability, causality, matching and physical cutoff remain open. The conditional  $Y^{3/2}$  weak-field branch and the rectangular- $T^3$  Casimir tensor are useful controlled results, but future predictive claims must remain downstream of the unresolved gates.

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